

Usability Study Report

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Usability study recording: <https://youtu.be/qf7BKTt6XL0>

Application scope

Application description:

PaletteForge is a full-stack web app that pairs a wireframe board with an accessibility toolkit, so that color decisions and accessibility checks happen in the same place instead of in separate tools at separate times. It is a single page React app backed by a Node/Express API and MongoDB, with session based authentication.

This study covers the Accessibility Toolkit vertical, which I built. It has four features behind a login: a Contrast Checker that computes the live WCAG 2.1 contrast ratio between any two colors and shows AA/AAA pass-fail badges for normal or large text; a Palette Builder that assigns colors to five fixed UI roles (background, primary, accent, text, border) and saves them as a named palette; an Audit & Report feature that runs a saved palette against a fixed set of role pairings and produces a saved pass/fail report; and Auto-Fix, which nudges a failing foreground color's lightness in HSL space until it meets its contrast target while preserving its hue. The Wireframe Studio half of the app was built by my project partner, Aishwarya Rajmohan and is covered by her own usability study.

Users - Target audience:

The toolkit is aimed at people who choose colors for interfaces but are not accessibility specialists, and who need a defensible answer rather than a gut feeling.

Maya, 24- junior frontend developer. Six months out of a bootcamp, first job at a 12 people startup. Comfortable in React and CSS. Has heard "WCAG" in code review but has never run a contrast check and does not know what AA means. Picks colors from a Figma file a designer handed her and assumes they're fine. Needs a tool that tells her whether a pairing passes without requiring her to already understand the standard.

Dev, 31- Developer. Ships small SaaS products alone, no designer. Received a support email from a user with low vision saying his dashboard text was unreadable, and wants to fix it without redesigning his brand. Cares specifically about keeping his existing colors recognizable, which is what Auto-Fix is for.

Ana, 28- UX designer at a small agency. Uses Figma daily and knows roughly what contrast ratios are. A recent government client contract made WCAG AA a deliverable requirement. She needs to check a whole palette at once and produce something she can attach to a handoff document, not check one pair at a time.

The common thread: all three are choosing colors for a real screen, all three need a pass/fail verdict they can act on, and none of them wants to compute relative luminance by hand.

Data description:

All data is stored in MongoDB via the official Node driver and is scoped per authenticated user. Three collections belong to this:

users- username and a bcrypt-hashed password. Never displayed, used only to authenticate and to scope every other record. Sessions are held in express-session cookies rather than in the document.

contrastPalettes- a user-named palette storing one hex value for each of the five UI roles (background, primary, accent, text, border), plus the owning user's ID. Displayed as labeled swatches in the Palette Builder and as options in the Audit & Report dropdown. This collection is also seeded with 1000+ synthetic records to test the audit feature at scale.

contrastReports- a saved audit result: a reference to the palette audited, and for each role pairing the two colors, the computed contrast ratio, and its AA/AAA pass-fail outcome for normal and large text. Displayed as the generated report table.

Two further collections, wireframes and extractedPalettes, belong to the Wireframe Studio vertical and are not exercised by this study.

Data displayed but not stored: live contrast ratios in the Contrast Checker, and Auto-Fix before/after results, both computed on the fly from `utils/wcag.js`.

Main tasks - use cases:

- **T1- Create an account and confirm the session holds.** Register a new account, reach the toolkit authenticated, reload the page and correctly report whether the session persisted, sign out, and sign back in with the same credentials.
- **T2- Determine whether a color pair is legible.** Given #7A7A7A as text on #FFFFFF as background, use the Contrast Checker to report the contrast ratio (4.29:1), report that it fails AA for normal-size body text, then locate the Large text checkbox and report that it passes AA for large text.
- **T3- Define and save a role-based palette.** In the Palette Builder, assign a chosen color to each of the five UI roles, give the palette a name, and save it so that it appears in the Saved palettes list.
- **T4- Audit a palette and identify what fails.** In Audit & Report, select the saved palette, generate a report, and correctly identify at least one specific role pairing that fails its contrast target.
- **T5- Repair a failing color without changing its hue.** Take a failing pairing from the report, get it into the app's contrast state, run Auto-Fix on the foreground, obtain a result that meets the contrast target, and articulate that the fixed color is the same hue at a different lightness.

Experiment

Participant 1

Demographics answers:

Question	Answer
Age range	23
Role / field	Student-Software Developer
Built a web page before?	Yes
WCAG familiarity	Little knowledge
Used a contrast tool before?	Yes
Comfortable with hex codes	No
Color vision / assistive tech	No
Browser / OS	Brave/Windows

Detailed notes

Notes for initial approach (intuitiveness)

- First-impression quote: “Interesting, would like to use the application more”
- Understood the app's purpose: Yes
- Reaction to the account/login requirement: Nothing said
- Named both top-level tabs unprompted: No
- Predicted what "Accessibility Toolkit" contained: Yes
- Sub-tabs that read as the same feature: No

Notes for T1- Authentication

- Outcome: success
- Time: 1min with passport explanation
- Timestamp: 5:30 - 6:30
- Assists: None
- Understanding: Completely understood
- What happened: Logged in, reloaded and logged out again
- Quotes: None

- Expectation violations: None observed
- Bugs:None

Notes for T2- Contrast Checker

- Outcome: Success
- Time: 2 mins with explanations
- Timestamp: 6:25 - 8:25
- Assists: None
- Understanding: Had an idea of what it is before the explanation
- Right ratio achieved: Yes
- Found the Large text checkbox: Yes
- What happened: The participant was able to explain what Contrast Checker is
- Bugs: None

Notes for T3- Palette Builder

- Outcome: Success with assists
- Time: 2 mins 15 secs
- Timestamp: 8:30 - 10:45
- Assists: Had to show the participant that a user can choose colors not only from the color spectrum but also with hex codes.
- Understanding: Had an idea of what it is before the explanation
- Color palette navigation: Yes
- What happened: The participant was able to explain what Palette builder is
- Bugs: None

Notes for T4- Audit & Report

- Outcome: Success
- Time: 1 min 6 secs
- Timestamp: 10:48 - 11:54
- Assists: None
- Could they explain why it matters: Not entirely, I had to explain the uses of Audit and report
- What happened: User generated report and wanted to know where the report would be useful
- Bugs: None

Notes for T5: Auto-Fix

- Outcome: Success with assist
- Time: 1 min 15 secs
- Timestamp: 11:55 - 13:10
- Assists: User could not understand this feature and I had to assist
- Can the user figure out fixed color: No, I had to assist

- What happened: The user auto fixed but didn't know where to see the fix
- Changes quoted: The user felt that the auto fix could have been in the Contrast Checker page
- Bugs: None

Notes for wrap-up probe

- One-sentence description of the toolkit: Cool website; Useful for frontend developers
- One thing they'd change: The user felt that the auto-fix could have been in the Contrast Checker page

Post-test questionnaire results (Likert scale 1-5)

	Item	Score	Participant's stated reason
1a	Effectiveness - T1	5	Simple and straightforward task
1b	Ease - T1	5	Easy
2a	Effectiveness - T2	5	The task was self explanatory
2b	Ease - T2	5	Simple and efficient
3a	Effectiveness - T3	4	Does its job
3b	Ease - T3	4	UI could be more bright
4a	Effectiveness - T4	5	Very descriptive report
4b	Ease - T4	5	Easy
5a	Effectiveness - T5	3	Placement could be better
5b	Ease - T5	3	Placement could be better
6a	Overall effectiveness	4	Pretty good job
6b	Overall ease	4	Easy

Participant 2

Demographics answers:

Question	Answer
Age range	23
Role / field	Student-Software Developer
Built a web page before?	Yes
WCAG familiarity	No knowledge
Used a contrast tool before?	Not used, but aware
Comfortable with hex codes	Has knowledge on what it is
Color vision / assistive tech	No
Browser / OS	Chrome/Windows

Detailed notes

Notes for initial approach (intuitiveness)

- First-impression quote: “Interesting, would like to use the application more”
- Understood the app's purpose: Yes
- Reaction to the account/login requirement: Nothing said
- Named both top-level tabs unprompted: No
- Predicted what "Accessibility Toolkit" contained: No
- Sub-tabs that read as the same feature: No

Notes for T1- Authentication

- Outcome: success
- Time: 1 min with render reload
- Timestamp: 19:45 - 20:48
- Assists: User was trying to sign in instead of logging in after creating an account
- Understanding: Completely understood
- What happened: Logged in, reloaded and logged out again
- Quotes: None
- Expectation violations: None observed
- Bugs: Render reload bug

Notes for T2- Contrast Checker

- Outcome: Success
- Time: 3 mins 30 secs with explanations
- Timestamp: 21:20 - 22:12 ; 27:00 - 28:31
- Assists: None
- Understanding: Had an idea of what it is before the explanation
- Right ratio achieved: Yes
- Found the Large text checkbox: Yes
- What happened: The participant was able to explain what Contrast Checker is
- Bugs: None

Notes for T3- Palette Builder

- Outcome: Success with assists
- Time: 1 mins 30 secs
- Timestamp: 28:33 - 30:00
- Assists: None.
- Understanding: Understood the concept but was unsure about the usage of this concept
- Color palette navigation: Yes
- What happened: The participant was able to explain what Palette builder is
- Bugs: None

Notes for T4- Audit & Report

- Outcome: Success with assist
- Time: 30:00 - 31:10
- Timestamp: 1 min 10 secs
- Assists: Had to assist with “Generate Report” button
- Could they explain why it matters: Yes, now they had a clear understanding of the previous concept-”Palette Builder” and why that is used to generate a report here.
- What happened: User generated report and had a clear understanding of it.
- Bugs: None

Notes for T5: Auto-Fix

- Outcome: Success
- Time: 1 min 30 secs
- Timestamp: 31:11 - 32:40
- Assists: Surprisingly, none
- Can the user figure out the fixed color: The user was able to navigate to “Contrast Checker” without any assistance to check the fix.
- What happened: The user auto fixed and knew where to see the fix
- Changes quoted: The user felt that the auto fix could have been in the Contrast Checker page
- Bugs: None

Notes for wrap-up probe

- One-sentence description of the toolkit: Nicely developed; Clean UI
- One thing they'd change: The user felt that the auto-fix could have been in the Contrast Checker page

Post-test questionnaire results (Likert scale 1-5)

	Item	Score	Participant's stated reason
1a	Effectiveness - T1	5	Smooth, no cookies based issue
1b	Ease - T1	5	Easy
2a	Effectiveness - T2	4	Missing the auto-fix option in this page
2b	Ease - T2	5	Simple and efficient
3a	Effectiveness - T3	5	UI was clean
3b	Ease - T3	5	Easy
4a	Effectiveness - T4	5	Very descriptive report
4b	Ease - T4	5	Easy
5a	Effectiveness - T5	4	By itself is really great
5b	Ease - T5	4	Easy
6a	Overall effectiveness	5	Pretty good job, No guidance needed
6b	Overall ease	5	Easy

Participant 3

Demographics answers:

Question	Answer
Age range	23
Role / field	Student-Software Developer
Built a web page before?	Yes
WCAG familiarity	No knowledge
Used a contrast tool before?	No
Comfortable with hex codes	No
Color vision / assistive tech	No
Browser / OS	Chrome/Mac

Detailed notes

Notes for initial approach (intuitiveness)

- First-impression quote: “I really like the idea of the project, will be useful if I were to design my room”
- Understood the app's purpose: Yes
- Reaction to the account/login requirement: Used a test login, then created a new account.
- Named both top-level tabs unprompted: No
- Predicted what "Accessibility Toolkit" contained: No
- Sub-tabs that read as the same feature: No

Notes for T1- Authentication

- Outcome: success
- Time: 1 min with passport authentication explanation
- Timestamp: 39:55 - 40:53
- Assists: User was trying to sign in instead of logging in after creating an account
- Understanding: Completely understood
- What happened: Logged in, reloaded and logged out again
- Quotes: None
- Expectation violations: None observed
- Bugs: None

Notes for T2- Contrast Checker

- Outcome: Success
- Time: 2 mins 6 secs with explanations
- Timestamp: 43:00 - 45:06
- Assists: None
- Understanding: Figured out what it is with my explanation
- Right ratio achieved: Yes
- Found the Large text checkbox: Yes
- What happened: The participant was able to explain what Contrast Checker is
- Bugs: None

Notes for T3- Palette Builder

- Outcome: Success
- Time: 1 mins 30 secs
- Timestamp: 45:09 - 46:40
- Assists: None.
- Understanding: Understood the concept but was unsure about the usage of this concept
- Color palette navigation: Yes
- What happened: The participant asked me for an explanation about this page before they could do anything.
- Bugs: None

Notes for T4- Audit & Report

- Outcome: Success with assist
- Time: 46:40 - 47:13
- Timestamp: Close to a min
- Assists: Had to assist with “Generate Report” button
- Could they explain why it matters: No, but it sounded like they got a hold of it
- What happened: User generated report and had a clear understanding of it.
- Bugs: None

Notes for T5: Auto-Fix

- Outcome: Success with assists
- Time: 1 min 30 secs
- Timestamp: 47:14 - 48:25
- Assists: There was a big pause so I had to interfere
- Can the user figure out the fixed color: The user was able to navigate to “Contrast Checker” without any assistance to check the fix.
- What happened: The user auto fixed and knew where to see the fix
- Changes quoted: The user felt that the auto fix could have been in the Contrast Checker page
- Bugs: None

Notes for wrap-up probe

- One-sentence description of the toolkit: Nicely developed; Insisted some changes to work with
- One thing they'd change: The user felt that the auto-fix could have been in the Contrast Checker page

Post-test questionnaire results (Likert scale 1-5)

	Item	Score	Participant's stated reason
1a	Effectiveness - T1	4	Insisted on implementing password protection, which I explained that I already have implemented on a smaller scale.
1b	Ease - T1	5	Easy
2a	Effectiveness - T2	5	No corrections
2b	Ease - T2	5	Simple and efficient
3a	Effectiveness - T3	5	Expected report on the same page
3b	Ease - T3	4	Expected report on the same page
4a	Effectiveness - T4	5	Very descriptive report
4b	Ease - T4	5	Easy
5a	Effectiveness - T5	4	By itself is really great
5b	Ease - T5	4	Easy
6a	Overall effectiveness	5	Pretty helpful for targeted users
6b	Overall ease	5	Easy

Cross-participant summary

Task outcomes

Task	P1	P2	P3
T1 Authentication	5	5	4
T2 Contrast Checker	5	4	5
T3 Palette Builder	4	5	4
T4 Audit & Report	5	5	5
T5 Auto-Fix	3	4	4

Prioritized list of issues and corresponding changes

Summary of the most crucial issues

The clearest result of the study was unanimous: all three participants independently said that Auto-Fix should be inside the Contrast Checker rather than as a separate tab. It was the answer every participant gave to "what one thing would you change" and it accounted for every low score in the study. Merging the two views was therefore the primary change made after testing.

Three smaller changes followed from the same sessions. Two of three participants needed help finding the Generate Report button, two of three could operate the Palette Builder without being able to say what a saved palette was for. Each was addressed with a small, targeted change. Three further observations were recorded but not implemented in this iteration and are documented below.

How priority was assigned

Priority	Assigned when
Must	Affected all three participants, or blocked a task
Should	Required an assist; affected two of three participants
Could	Affected one participant, or was recoverable and non-blocking
Would	Feature request beyond the scope of this iteration

- **Issue 1:** All three participants said Auto-Fix should not be a separate tab. It was the unanimous answer to the wrap-up question, and it produced the lowest scores in the study: P1 rated T5 effectiveness and ease 3 and 3, citing placement, and P2 marked T2 effectiveness down to a 4 specifically because the auto-fix option was absent from the Contrast Checker page. Two of three participants also could not locate the fix result without help. P1 ran the fix but did not know where to look for the output, and P3 paused long enough that the moderator intervened. The root cause is that Auto-Fix has no inputs of its own and silently reads whatever colour pair the Contrast Checker holds, so the two features are already coupled in code while being separated in the interface. (P1 @ 11:55–13:10, P2 @ 31:11–32:40, P3 @ 47:14–48:25)

Change: Merge Auto-Fix into the Contrast Checker view. When a pairing fails, show an "Auto-fix foreground" action directly beneath the ratio and badges, with the fixed hex value and its new ratio displayed in place. Remove Auto-Fix from the sub-tab bar, taking the toolkit from four tabs to three.

Priority: Must. 3 of 3 participants and the only change every participant requested unprompted.

- **Issue2:** Two of three participants needed assistance to find the Generate Report button. The button is disabled until a palette is chosen from the dropdown, but nothing communicates that dependency, so the greyed-out control read as non-functional rather than as awaiting input. (P2 @ 30:00–31:10, P3 @ 46:40–47:13)

Change: Add helper text beside the disabled button reading "Select a palette to enable" plus a title attribute explaining the disabled state on hover.

Priority: Should. 2 of 3 participants required an assist on an otherwise trivial step.

- **Issue3:** Two of three participants could operate the Palette Builder but could not explain what it was for. P2 understood the mechanics but was unsure of the purpose, grasping it only after reaching Audit & Report. P3 asked for an explanation of the page before attempting anything. Nothing on the page indicates what a saved palette is subsequently used for. (P2 @ 28:33–30:00, P3 @ 45:09–46:40)

Change: Add a one-line description beneath the Palette Builder heading stating that saved palettes are audited for contrast in the next step.

Priority: Should. 2 of 3 participants, and it weakens the intended flow between two features.

- **Issue 4:** One participant asked whether passwords were protected and suggested stronger password handling, costing T1 a point on effectiveness. Passwords are already hashed with bcrypt, but nothing in the interface communicates that, and no minimum password requirements are enforced at registration. (P3, Likert 1a)

Change: Enforce a minimum password length at registration with inline feedback, and note on the form that passwords are stored hashed.

Priority: Would. a single participant, and the underlying security concern is already addressed by bcrypt hashing. The request was for additional strength enforcement, which is outside the scope of this iteration.